

Hybrid Recommender Evaluation Report

Eligible users with next-purchase holdout: 322

Ranking rows: 26877

Average candidate pool size: 83.47

Signal sources:

1. Collaborative Filtering score
2. Embedding similarity score
3. Price sensitivity match
4. Style preference match
5. Historical CTR (simulated)

Scoring function:

$\text{hybrid_signal} = w_{\text{cf}} \cdot \text{cf_score} + w_{\text{embedding}} \cdot \text{gated_embedding_similarity} + w_{\text{price}} \cdot \text{price_sensitivity_match} + w_{\text{style}} \cdot \text{style_affinity_score} + w_{\text{ctr}} \cdot \text{historical_ctr}$

Best signal weights:

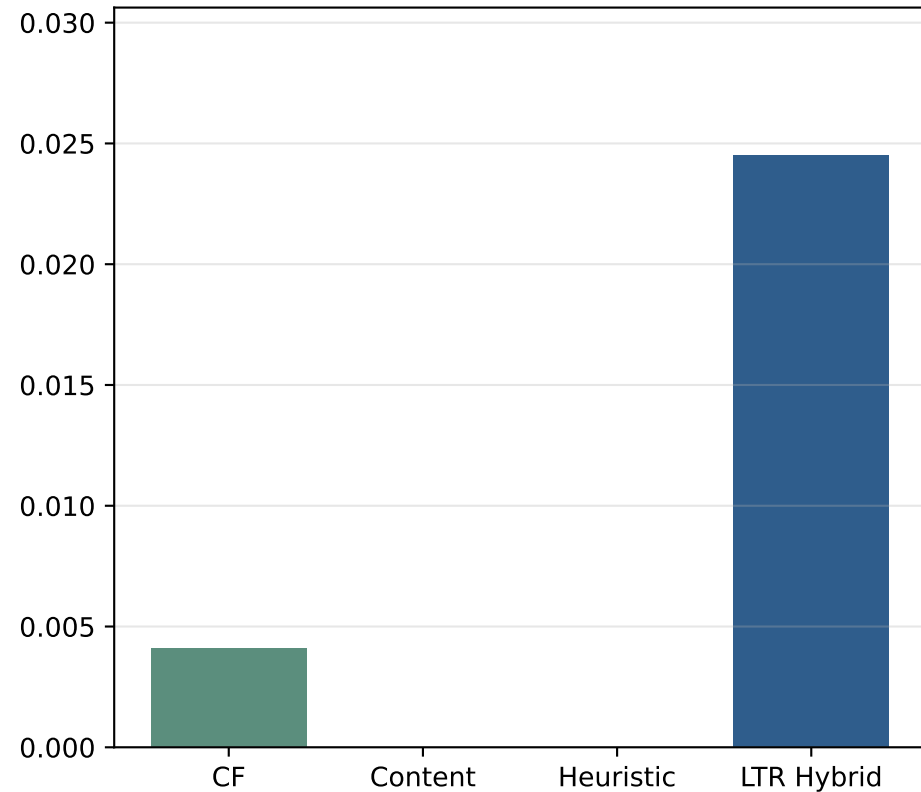
- cf: 0.165
- embedding: 0.075
- price: 0.132
- style: 0.457
- ctr: 0.172

Similarity threshold: 0.232

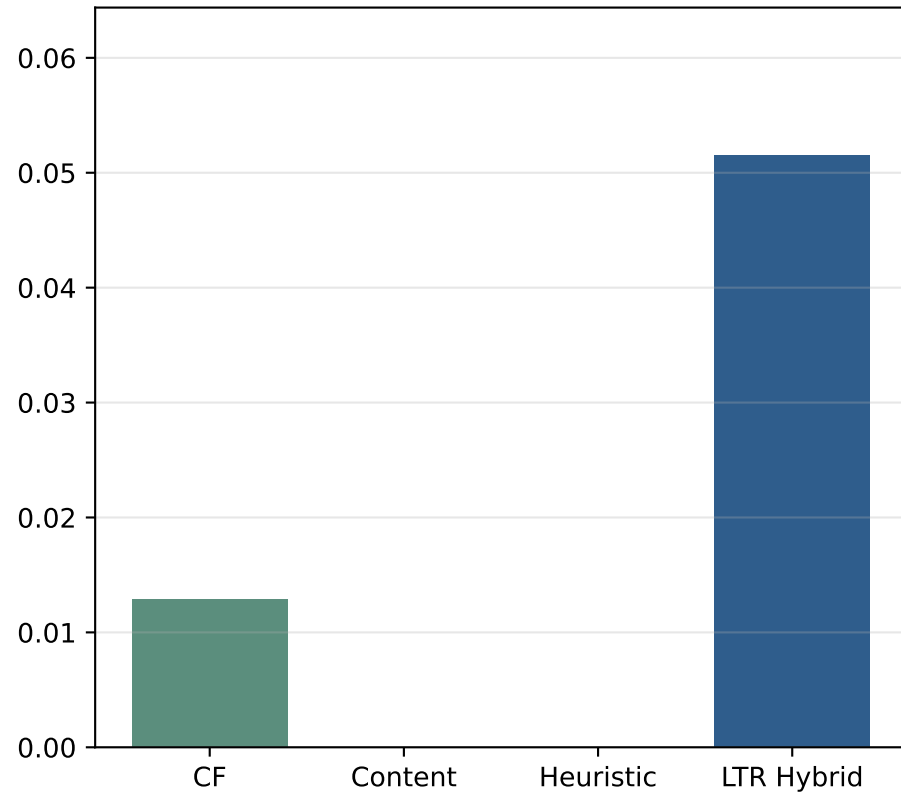
Headline test-set lift vs strongest baseline:

- Precision@5: +0.0204 (+497.56%)
- NDCG@5: +0.0386 (+299.22%)

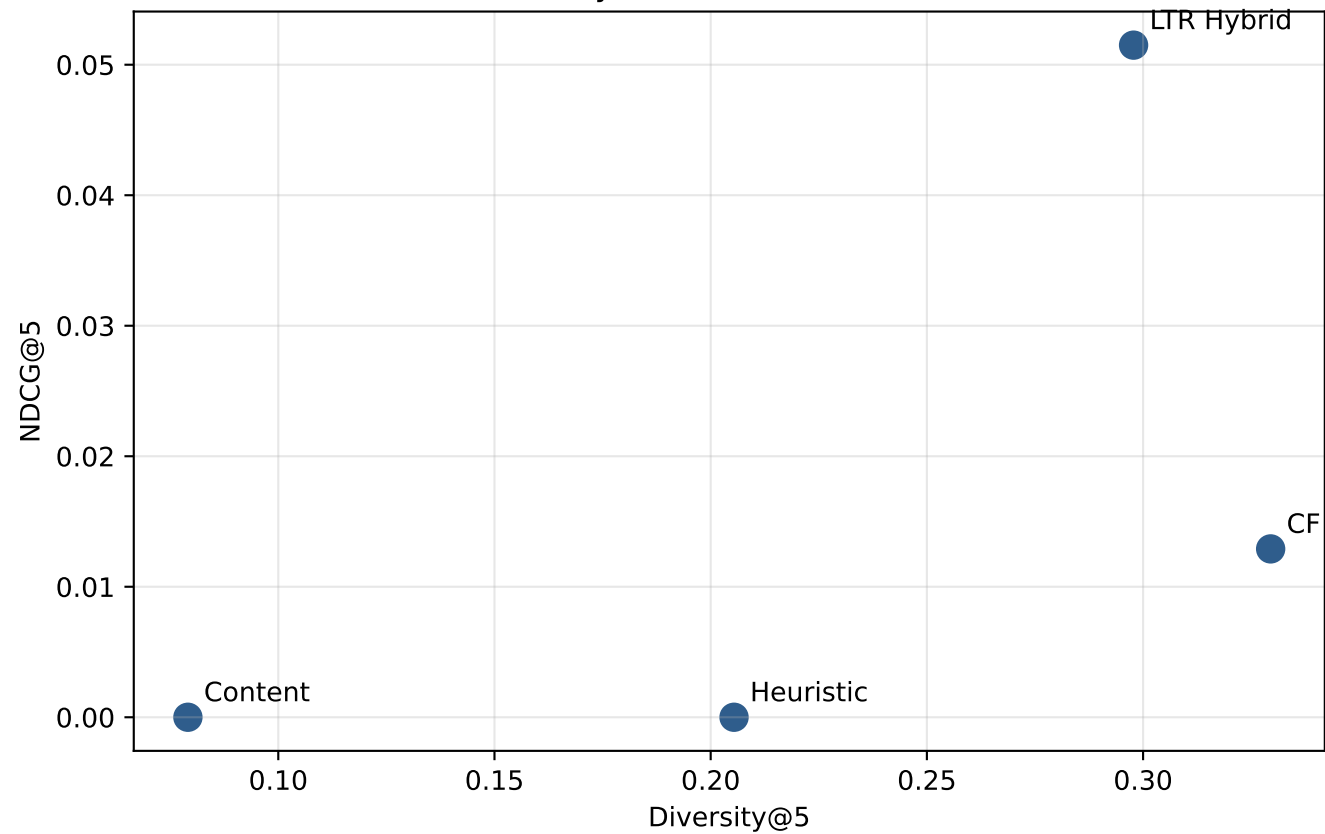
Precision@5



NDCG@5



Diversity vs Relevance Trade-off



LTR Feature Importance

